

Haoyang Zheng

PH.D. CANDIDATE & APPLICANT

☎ (765) 413-7189 | ✉ zheng528@purdue.edu | 🏠 haoyangzheng.github.io | 🌐 github.com/haoyangzheng1996 | 📄 haoyangzheng | 🎓 Scholar

Education

Purdue University, College of Engineering

Ph.D. in Mechanical Engineering (advised by Prof. Guang Lin, GPA 4.0/4.0)

West Lafayette, IN

Jun. 2021 - May. 2025 (expected)

Purdue University, College of Engineering

M.S. in Mechanical Engineering (GPA 3.8/4.0)

West Lafayette, IN

Sep. 2019 - May. 2021

Southwest University, School of Computer and Information Science

B.Eng. in Automation (advised by Prof. Yong Deng, Rank 1/92)

Chongqing, China

Sep. 2014 - Jun. 2018

Professional Experience

Argonne National Laboratory, Mathematics and Computer Science Division

Lemont, IL

Givens Associate

May. 2023 - Jul. 2023

- Investigated suitable hyperparameters for **proximal policy optimization (PPO)** models through centralized Bayesian optimization search.
- Integrate the DeepHyper framework with the **MPI program** to achieve parallel computing and improve exploration.

Conference

- [1] **Haoyang Zheng**, Wei Deng*, Christian Moya, Guang Lin*, “Accelerating Approximate Thompson Sampling with Underdamped Langevin Monte Carlo”, *AISTATS 2024*;
- [2] **Haoyang Zheng**, Hengrong Du, Qi Feng, Wei Deng*, Guang Lin*, “Constrained Exploration via Reflected Replica Exchange Stochastic Gradient Langevin Dynamics”, *ICML 2024*.
- [3] **Haoyang Zheng**, Ruqi Zhang*, Guang Lin*, “Exploring Non-Convex Discrete Energy Landscapes: A Langevin-Like Sampler with Replica Exchange”, submitted to *ICLR 2025*.

Journal

- [1] **Haoyang Zheng**, Yao Huang, Ziyang Huang, Wenrui Hao, Guang Lin*, “HomPINNs: Homotopy Physics-Informed Neural Networks for Solving the Inverse Problems of Nonlinear Differential Equations with Multiple Solutions”, *Journal of Computational Physics* (2024);
- [2] **Haoyang Zheng**, Jeffrey R. Petrella, P. Murali Doraiswamy, Guang Lin*, Wenrui Hao, “Data-Driven Causal Model Discovery and Personalized Prediction in Alzheimer’s Disease”, *NPJ Digital Medicine* (2022);
- [3] **Haoyang Zheng**, Ziyang Huang, Guang Lin*, “A Physics-Constrained Neural Network for Multiphase Flows”, *Physics of Fluids* (2022);
- [4] **Haoyang Zheng**, Yong Deng*, Yong Hu, “Fuzzy Evidential Influence Diagram and Its Evaluation Algorithm”, *Knowledge-Based Systems* (2017);
- [5] **Haoyang Zheng**, Yong Deng*, “Evaluation Method Based on Fuzzy Relations Between Dempster-Shafer Belief Structure”, *International Journal of Intelligent Systems* (2018);
- [6] Tian Bian, **Haoyang Zheng**, Yong Deng*, “Failure Mode and Effect Analysis Based on D Numbers and Topsis”, *Quality and Reliability Engineering International* (2018);
- [7] Likang Yin, **Haoyang Zheng**, Tian Bian, Yong Deng*, “An Evidential Link Prediction Method and Link Predictability Based on Shannon Entropy”, *Physica A* (2017);
- [8] **Haoyang Zheng**, Guang Lin*, “LES-SINDy: Laplace-Enhanced Sparse Identification of Nonlinear Dynamical Systems”, submitted to *Nature Computational Science*.

Talks

[1] On Sampling Tasks with Langevin Dynamics, Brown University, July 2024.
Host: George Em Karniadakis.

Achievements

SELECTED AWARDS

2024	Student Travel Awards, 2024 Mathematical and Scientific Foundations of Deep Learning Annual Meeting	New York, NY
2024	Student Travel Awards, 2024 SIAM Conference on Mathematics of Data Science (MDS24)	Atlanta, GA
2024	Student Travel Awards, Nonlocality: Challenges in Modeling and Simulation	Providence, RI
2023	Student Travel Awards, 2023 Mathematical and Scientific Foundations of Deep Learning Annual Meeting	New York, NY
2018	Pacemaker to Technological Innovation in Chongqing, Awarded 10 college students every two years	Chongqing, China
2018	Outstanding undergraduates in Chongqing, Awarded 1% of all college undergraduate students every year	Chongqing, China
2015	China National Scholarship, Awarded 1% of all undergraduates in China	China

SELECTED HONORS

2017	Finalist (0.5%), Interdisciplinary Contest in Modeling	China
2016	Special prize (2/3568), International Mathematical Contest in Modeling	China

Teaching Experience

Purdue University, School of Mechanical Engineering West Lafayette, IN
Teaching Assistant Aug. 2019 - May. 2021

- Guided students through **hands-on lab tasks** in ME375 (Measurement And Control Systems) and ME475 (Automatic Control Systems), ensuring understanding of key principles and safe practice.
- Conducted regular **office hours** to provide individualized support and address questions in homework and lab sections.
- Collaborated with instructors to **organize and execute in-class robot competitions**.

Service

Conference Reviewer ICLR, ICML, NeurIPS, AISTATS
Journal Reviewer Scientific Reports, Journal of Computing and Information Science in Engineering, Journal of Supercomputing, Knowledge and Information Systems

Skills

Programming Python (TensorFlow, PyTorch), MATLAB, R, C, Java
Research skills LaTeX, Origin Lab, EndNote, Visio, Notion, Obsidian